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DATect: Per-Site Machine Learning Forecasting of Domoic Acid Along the Pacific Northwest Coast

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Abstract: Domoic acid (DA) contamination of shellfish is a recurring public health crisis along the U.S. Pacific Northwest coast, yet no quantitative forecasting system has been published for the region. We present DATect, a one-week-ahead site-specific DA forecasting system for 10 monitoring sites spanning 600 km of Washington and Oregon coast, trained on 21 years (2003–2023) of satellite, climate, hydrological, and biological data. DATect couples a per-site XGBoost + Random Forest ensemble with a transition-focused binary classifier; all per-site configuration is selected by leak-free chronological cross-validation on pre-2022 data, making the 2022–2023 temporal holdout fully unbiased. On that holdout (every real DA measurement, $N = 404$), the regression ensemble achieves pooled $R^2 = 0.485$ [95% CI: 0.330, 0.604], MAE = 6.76 ¯g/g, and spike recall = 0.857 at the 20 ¯g/g regulatory threshold, with 9 of 10 sites measurably positive. Combining the regression with the transition classifier raises hybrid-alert recall to 87.6% on the full retrospective panel. DATect is deployed as a publicly accessible web dashboard.

Keywords: domoic acid; harmful algal blooms; machine learning; ensemble forecasting; *Pseudo-nitzschia*; Pacific Northwest; XGBoost; data leakage prevention

1. Introduction

1.1. Domoic Acid and Pseudo-nitzschia Blooms

Domoic acid (DA) is a potent neurotoxin produced by certain species of the marine diatom genus *Pseudo-nitzschia* and is the causative agent of amnesic shellfish poisoning [1–3]. Since the first documented global outbreak in 1987, DA contamination has become a persistent threat along the U.S. Pacific coast, with severe consequences for human health, coastal economies, and marine wildlife [2,4,5]. The 2015 marine heatwave illustrates the scale of this risk: DA concentrations in razor clams exceeded the federal safety limit by over 700%, causing tens of millions of dollars in fishery losses [6–8].

Forecasting DA is difficult because toxin production is only weakly coupled to bloom magnitude. DA production depends strongly on physiological stress—particularly nutrient limitation—so a large *Pseudo-nitzschia* bloom may be weakly toxic while a smaller bloom generates dangerous contamination [9–11]. In the Pacific Northwest (PNW), this biology interacts with coastal upwelling, climate variability, and retention zones such as the Juan de Fuca eddy and Heceta Bank, creating a multiscale nonlinear system with strong spatial heterogeneity [12–15].

1.2. Existing Monitoring and Forecasting in the Pacific Northwest

Current DA risk management in the PNW is reactive: beaches are closed only after measured toxin levels exceed the FDA regulatory limit of 20 ¯g/g [16], giving managers no advance warning. The primary forecasting tool is the NOAA Pacific Northwest HAB Bulletin [16], issued biweekly to monthly during peak bloom season. The Bulletin synthesizes environmental and biological data into qualitative risk categories (low/medium/high)

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that provide strategic planning value [17] but do not produce numerical DA predictions and operate at seasonal rather than weekly resolution. Complementary regional resources include Oregon's phytoplankton monitoring program [18] and the SoundToxins network [19], which extend field observations of harmful phytoplankton and toxins, together with the LiveOcean circulation model [20], which supplies numerically simulated coastal and estuarine circulation fields; none of these systems produce numerical DA forecasts. The California C-HARM system [21] provides bloom-probability nowcasts for California but does not extend to Washington or Oregon (Table 1).

A critical gap thus exists: **no quantitative DA concentration forecasting system has been developed for the Pacific Northwest**. All existing monitoring is reactive, and the sole forecasting tool provides qualitative risk categories, not numerical toxin predictions.

1.3. Problem Statement and Contributions

This paper addresses the absence of quantitative DA forecasting in the PNW by developing DATect, the first system to produce site-specific, numerical DA concentration predictions for this region. We formalize the prediction task as follows: given environmental features available at time $t - 7$ days (the "anchor date"), predict the DA concentration (¯g/g) at time t for a specific monitoring site s among 10 Pacific Coast locations.

Two evaluation targets matter operationally. The first is magnitude forecasting: how accurately can a system predict DA concentration one week ahead? The second is transition detection: can the system anticipate below-to-above-threshold crossings at 20 ¯g/g, the events that trigger harvest closures? DATect is evaluated on both targets because a model can achieve low average regression error while still missing the closure-relevant transitions that matter most to managers.

This task presents four compounding challenges:

1. **Sparse, irregular target variable**: DA measurements are collected approximately weekly but with frequent gaps, missingness, and inconsistent timing across sites.
2. **High spatial variability**: The 10 monitoring sites span 5° of latitude (∼600 km), with dramatically different oceanographic regimes, bloom histories, and DA dynamics.
3. **Extreme value distribution**: DA measurements are dominated by zeros and low values, with rare but consequential spikes exceeding 100 ¯g/g.
4. **Strict temporal integrity requirements**: Operational forecasting demands that no future information leak into predictions, requiring rigorous validation protocols.

This paper makes five main contributions:

1. **Quantitative PNW DA forecasting under a strict 3-window chronological protocol**: DATect provides the first site-specific, one-week-ahead DA concentration forecasts for Washington and Oregon, using 21 years of data across 10 monitoring sites. All configuration decisions (per-site ensemble weights, prediction clipping) are selected via leak-free chronological cross-validation on pre-2022 data only, making the 2022–2023 temporal holdout a fully unbiased generalization estimate.
2. **Per-site modeling on sparse, irregular records**: Per-site XGBoost or Random Forest configurations, interpolated training, and raw-only persistence features provide a practical approach to sparse monitoring data while keeping evaluation tied to real observations.
3. **Transition-focused early warning**: A dedicated binary classifier trained with a safe-baseline filter provides advance warning for below-to-above-threshold crossings that trigger harvest closures, complementing the regression forecast rather than replacing it (Section 4.5).
4. **Evidence that data quality is the binding constraint**: The Washington–Oregon performance split shows that monitoring density and regularity, not model complexity, are the primary limits on forecast skill in the current regional record.
5. **Fully reproducible tuning workflow**: All tunable hyperparameters and chronological evaluation windows are stored in a single source-of-truth JSON file (`config/tuned_hyperp`

with per-key provenance, enabling reviewers and future contributors to re-run the entire selection workflow and reproduce the headline numbers end-to-end.

2. Related Work

2.1. Quantitative Forecasting of Domoic Acid and Pseudo-nitzschia

Despite decades of DA monitoring and multiple qualitative forecasting systems, quantitative computational approaches specifically targeting DA or *Pseudo-nitzschia* remain limited in number, geographic scope, and temporal extent [22,23].

In the Galician Rías Baixas region of northwest Spain, Aláez et al. [24] applied Random Forest and AdaBoost classifiers to predict *Pseudo-nitzschia* bloom events, achieving approximately 85% classification accuracy. Their study covered five monitoring stations over approximately a decade of data, targeting bloom occurrence (binary classification) rather than DA concentration. Cusack et al. [25] modeled *Pseudo-nitzschia* events off southwest Ireland using environmental predictors, demonstrating the potential of empirical models for bloom forecasting in European waters. Additional studies have documented *Pseudo-nitzschia* dynamics and DA contamination in new regions, including the Gulf of Maine [26] and southern Florida [27], reflecting the global expansion of DA monitoring efforts.

In the California Current, Lane et al. [28] developed a logistic regression model for predicting toxigenic *Pseudo-nitzschia* blooms in Monterey Bay using 8.3 years of monitoring data, achieving $\geq$75% predictive accuracy with chlorophyll *a* and silicic acid as primary predictors.

The closest methodological parallel to DATect is the deep learning system developed by Grasso et al. [29] for forecasting paralytic shellfish toxin (PST) levels in coastal Maine. Using a convolutional neural network (CNN) trained on chemical data from routine monitoring sites, their system forecasts PST levels on a weekly basis with over 95% prediction reliability. However, this system targets a different toxin (PST from *Alexandrium catenella*, not DA from *Pseudo-nitzschia*), covers a smaller geographic area (approximately 10 sites along the Maine coast), and uses approximately five years of data.

The California Harmful Algae Risk Mapping (C-HARM) system [21] provides satellite-based nowcasts of *Pseudo-nitzschia* bloom probability and DA risk along the California coast. C-HARM integrates satellite imagery with ocean model output to produce gridded risk maps but operates as a nowcast (current conditions) rather than a forecast and does not extend to Washington or Oregon. Moreno et al. [30] developed a mechanistic process-based model of *Pseudo-nitzschia* and DA production for the California Current, representing a complementary approach to data-driven methods. Most recently, Brunson et al. [31] demonstrated molecular forecasting of DA using gene expression (dabA and sit1 coexpression) as a one-week-ahead predictor during a bloom in Monterey Bay, representing an emerging genomics-based approach that could eventually complement satellite- and ML-based systems like DATect.

2.2. Machine Learning Methods for Harmful Algal Blooms

Across HAB forecasting more broadly, tree ensembles, linear models, and deep sequence models have all been applied to heterogeneous environmental data [32–36]. For DATect, the most relevant lesson is not that one model family dominates universally, but that coastal forecasting methods must tolerate missingness, nonlinear interactions, and mixed satellite/climate/biological covariates. Gradient boosting and Random Forests therefore provide natural nonlinear competitors [37–39], while linear models remain useful baselines for testing whether the gains come from model flexibility or from feature engineering and site-specific design.

2.3. Data Leakage in Environmental Machine Learning

A growing body of work has highlighted the pervasive problem of data leakage in machine learning applications to environmental science. Kapoor and Narayanan [40] demonstrated that data leakage affects a substantial fraction of published ML-based science papers, leading to inflated performance claims and irreproducible results. In temporal prediction tasks such as HAB forecasting, common sources of leakage include random train-test splits that violate temporal ordering, use of features computed from future data, and shared preprocessing across train and test sets [41–43]. Proper out-of-sample evaluation for temporal data requires rolling-origin or expanding-window approaches rather than random holdouts [44], and spatial autocorrelation in environmental data further complicates validation design [45].

DATect addresses the leakage problem explicitly through nine structural leakage-prevention mechanisms described in Section 3.4.

2.4. Comparison of Existing Systems

Table 1. Comparison of existing HAB forecasting systems with DATect. DATect provides the longest temporal record (21 years) and broadest geographic span (∼600 km) of any published quantitative DA/*Pseudo-nitzschia* ML forecasting study.

System	Region	Sites	Period	Method	Target	Horizon
PNW HAB Bulletin [16]	WA/OR	∼20	2017–	Qual. expert	DA risk (categ.)	1–2 wk
Maine PST [29]	Maine	∼10	∼5 yr	CNN	PST (not DA)	1 wk
Galician RF [24]	NW Spain	∼5	∼10 yr	RF/AdaBoost	PN bloom (bin.)	—
C-HARM [21]	California	Grid	Ongoing	Sat.+model	PN prob.	Nowcast
Monterey Bay LR [28]	California	1	∼8 yr	Logistic reg.	PN bloom (bin.)	—
DATect (this study)	**WA/OR**	**10**	**21 yr**	**XGB+RF**	**DA conc.+categ.**	**1 wk**

3. Methodology

3.1. Study Area and Data Collection

3.1.1. Monitoring Sites

DATect forecasts DA concentrations at 10 Pacific Coast monitoring sites spanning from Gold Beach, Oregon (42.38° N) to Kalaloch, Washington (47.59° N), a distance of approximately 600 km along the coastline (Figure 1). Sites were selected for their comprehensive historical datasets, consistent sampling over a multi-year period, and spatial representation across the Washington–Oregon coast. Pacific razor clams (*Siliqua patula*) are sampled at all sites. As filter-feeding bivalves with slow DA depuration (half-life of weeks to months), razor clams continuously accumulate toxin during bloom events, providing an integrated measure of recent DA exposure at each site [16,46,47].

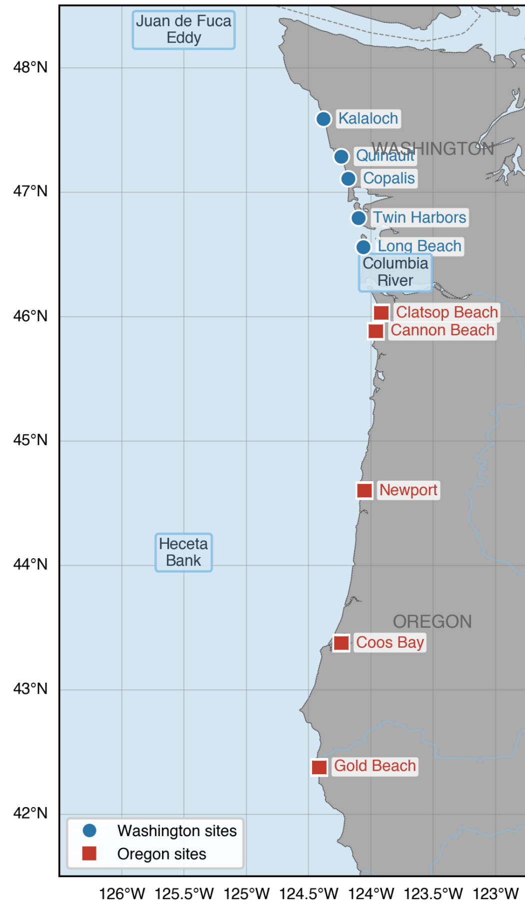

Figure 1. Study area showing the 10 DATect monitoring sites along the Pacific Northwest coast of Washington and Oregon. Sites span approximately 5° of latitude (∼600 km). The Juan de Fuca eddy region and Heceta Bank, two key bloom retention zones, are indicated.

Table 2 lists the 10 monitoring sites with their coordinates and the number of validation samples available at each site.

Table 2. DATect monitoring sites, geographic coordinates, and validation sample sizes.

Site	State	Latitude (°N)	Longitude (°W)	N (validation)
Kalaloch	WA	47.586	124.379	235
Quinault	WA	47.284	124.236	181
Copalis	WA	47.106	124.181	277
Twin Harbors	WA	46.792	124.100	225
Long Beach	WA	46.558	124.061	209
Clatsop Beach	OR	46.029	123.917	354
Cannon Beach	OR	45.882	123.959	116
Newport	OR	44.600	124.050	218
Coos Bay	OR	43.376	124.237	109
Gold Beach	OR	42.377	124.414	257
Total validation samples				**2,181**

3.1.2. Data Sources

DATect integrates seven categories of environmental and biological data spanning 21 years (2003–2023), summarized in Table 3. All data are aligned to a common weekly temporal resolution using Monday-anchored ISO weeks. The dataset is constructed by an automated pipeline that downloads, processes, and integrates all data sources into a unified analysis-ready file.

Table 3. Data sources used in DATect, with temporal resolution, period of record, and operational availability delays enforced during forecasting.

Source	Variables	Resolution	Period	Delay
MODIS-Aqua (CoastWatch)	Chl-*a*, SST, PAR, FLH, K490	8-day, ∼4 km	2003–2023	7 d
MODIS anomalies	Chl-*a* anom., SST anom.	Monthly, ∼4 km	2003–2023	2 mo
NOAA ERDDAP	PDO, ONI	Monthly	2003–2023	2 mo
NOAA ERDDAP	BEUTI	Daily	2003–2023	7 d
USGS NWIS	Columbia R. discharge	Daily	2003–2023	7 d
WDOH / ODFW	DA concentrations	Irregular (∼wk)	2003–2023	—
WDOH / ODFW	*Pseudo-nitzschia* cells	Irregular	2003–2023	—

Satellite data from MODIS-Aqua [48] are retrieved as 8-day composites from the NOAA CoastWatch ERDDAP server for each site's local bounding box. The use of 8-day composites mitigates frequent cloud cover in the PNW. A 7-day processing buffer is enforced to simulate operational satellite data availability delays.

Climate indices include the Pacific Decadal Oscillation (PDO) [49], the Oceanic Niño Index (ONI; NOAA operational index based on Niño 3.4 sea surface temperature anomalies in ERSSTv5) [50], and the Biologically Effective Upwelling Transport Index (BEUTI) [15]. BEUTI is provided at discrete 1° latitude bands; site-specific values are obtained via inverse distance weighting (IDW) to each site's exact latitude. PDO and ONI are subject to a 2-month reporting delay; BEUTI uses a 7-day buffer. Columbia River discharge from the USGS gauge at The Dalles proxies freshwater input to coastal salinity and nutrient dynamics, with the strongest signal at northern Washington sites.

DA measurements and *Pseudo-nitzschia* cell counts were obtained through public records requests to WDOH, WDFW, ODFW, and the Olympic Region Harmful Algal Bloom (ORHAB) Partnership. Measurements reported as below the detection limit ($<$ 1 ¯g/g) are imputed as 0.5 ¯g/g (half the detection threshold), a standard practice for left-censored environmental data [51]. These biological measurements are irregularly spaced, reflecting operational sampling schedules that intensify during suspected bloom events and relax during quiescent periods.

3.2. Data Engineering

3.2.1. Weekly Aggregation and Gap Handling

All data are aggregated to Monday-aligned ISO weeks. DA concentrations within each week are aggregated using the *maximum* operation—a conservative choice that preserves worst-case readings for public health relevance. When multiple measurements exist within a single week (common during active bloom monitoring), the maximum ensures that the highest detected toxin level is retained.

Gaps in the DA and *Pseudo-nitzschia* time series are filled using *exponential decay* interpolation. The key reason to prefer this over linear interpolation is temporal integrity: linear interpolation between two measurements requires the *future* measurement to compute intermediate values, introducing data leakage into the gap-filled training targets. Exponential decay propagates information only forward from the last known observation:

$$\hat{y}(t + \Delta t) = y(t) \cdot e^{-\lambda \Delta t} \tag{1}$$

where $y(t)$ is the last observed value, Δt is the gap duration in weeks, and λ is the species-specific decay rate ($\lambda_{\mathrm{DA}} = 0.2$ wk^{-1}, half-life $\approx$ 3.5 weeks, max gap 2 weeks; $\lambda_{\mathrm{PN}} =$

0.3 wk^{-1}, max gap 4 weeks). The exponential decay model also reflects the biological reality of DA depuration in razor clam tissue [46,47]. Gaps exceeding these thresholds are filled with zero, assuming non-detection.

Critical distinction: augmented training, raw-only evaluation. Exponential decay interpolation serves two purposes: (1) constructing the *environmental feature grid*—the weekly time series from which satellite, climate, and *Pseudo-nitzschia* features are drawn, and (2) *augmenting the training set* with gap-filled DA values. Training samples include both raw measurements and gap-filled DA values, providing approximately 5× more training data per site than raw-only training. Where a real measurement exists, it takes priority over the gap-filled value. This augmentation is controlled by the `USE_INTERPOLATED_TRAINING` configuration flag.

Critically, *test* samples remain restricted to weeks with actual raw DA measurements, ensuring all evaluation metrics reflect predictions of real toxin concentrations. Because the gap-filled values use forward-only exponential decay, they do not introduce temporal leakage into the training set. However, persistence features (`last_observed_da_raw`, `weeks_since_last_spike`) and the naïve persistence baseline are computed exclusively from real observations. These features should reflect what a coastal manager would actually *know* from field sampling, not synthetic estimates. A `_is_interpolated` marker column tracks which training rows use gap-filled targets, enabling this selective filtering. This design provides the statistical benefits of a larger training set while ensuring that persistence signals and evaluation metrics reflect actual field measurements.

3.2.2. Derived Environmental Features

Six derived features encoding domain knowledge—including a marine heatwave flag, BEUTI nonlinear terms, PDO–ONI phase coherence, *Pseudo-nitzschia* transforms, fluorescence efficiency, and a K490 quadratic term—were explored during development (Appendix F). Individual feature ablation (Table A8) confirmed that all produce negligible impact ($|\Delta R^2| \leq 0.005$), and all have been removed from the final pipeline. Similarly, several satellite products (chlorophyll-*a*, PAR, K490, chlorophyll anomaly) and site coordinates are ingested for ablation and audit but dropped before final model fitting, after confirming negligible marginal value (Appendix F).

3.2.3. Persistence Features, Lag Features, and Rolling Statistics

DATect also uses *observation-order lag features*—the Nth most recent raw observation strictly before the test date—rather than fixed time-grid shifts (Appendix F). Ablation shows these are largely redundant with the persistence features at the one-week horizon ($\Delta R^2 = -0.001$; Section 4.6).

Two persistence features are computed from raw DA observations:

- `last_observed_da_raw`: Forward-fill of the most recent DA measurement
- `weeks_since_last_spike`: Time since last DA > 20¯g/g event

Rolling statistics (mean, standard deviation, and maximum) are computed over windows of 4, 8, and 12 weeks for each site. Rolling means for 8- and 12-week windows and all rolling standard deviation features have negligible importance and are dropped; only the 4-week rolling mean/maximum and 8-/12-week maximum carry meaningful signal. To prevent the current week's DA from leaking into its own features, rolling statistics are computed on a `shift(1)` version of the persistence series—that is, the rolling window begins from the *prior* week's forward-filled DA value, not the current week's. This one-row offset ensures that a test point's rolling features reflect only DA history available *before* the prediction date, even if the test point itself corresponds to a raw measurement week.

3.2.4. Temporal Features and Preprocessing

Cyclical temporal features (sine/cosine encodings of day-of-year and month) capture seasonal bloom patterns without year-boundary discontinuities. Additional temporal encodings (week-of-year, bloom season flag, linear trend) were explored but confirmed negligible and removed (Appendix F). All numeric features undergo median imputation for missing values followed by min-max scaling to $[0, 1]$ using scikit-learn [52].

3.3. Model Architecture

3.3.1. Two-Model ML Ensemble

DATect employs a per-site model selection framework, formalized as a weighted two-model ensemble:

$$\hat{y} = w_{\mathrm{XGB}} \cdot \hat{y}_{\mathrm{XGB}} + w_{\mathrm{RF}} \cdot \hat{y}_{\mathrm{RF}} \tag{2}$$

where $w_{\mathrm{XGB}} + w_{\mathrm{RF}} = 1$ and the weights are site-specific (Table A1 in Appendix A). In the current configuration, all sites resulted in single-model assignments ($w \in \{0, 1\}$) based on development-set performance, making this effectively a per-site model selection between XGBoost and Random Forest. Blended weights (e.g., 50/50 or 60/40) were tested empirically but produced no improvement over single-model selection on either the held-out or temporal holdout evaluations ($\Delta R^2 \leq -0.004$); the ensemble formulation is retained for generality. Naïve persistence is computed for every prediction and reported as a standalone external baseline (Section 4.1), enabling a direct assessment of how much skill the ML models add beyond simple autocorrelation.

XGBoost [38]: A gradient boosted tree model using the histogram-based training method (`tree_method=hist`) with global base hyperparameters ($n_{\mathrm{estimators}} = 400$, max depth $= 6$, learning rate $= 0.05$, subsample $= 0.85$, colsample_bytree $= 0.85$, colsample_bylevel $= 0.8$, $\alpha = 0.1$, $\lambda = 1.0$, $\gamma = 0.1$, min_child_weight $= 3$) and per-site overrides (Appendix D). During retrospective evaluation, per-anchor hyperparameter selection uses a calibration fraction of 30% of training data (maximum 20 rows), searching over a site-specific candidate grid of 1–3 configurations. In real-time deployment, per-site hyperparameters are used directly without per-anchor tuning; this simplification is justified because the perturbation study (Appendix K) shows that XGBoost performance is insensitive to small hyperparameter variations within the tested range. XGBoost's native handling of missing values and its L1/L2 regularization make it well-suited for the noisy, gap-filled environmental feature space.

Random Forest [39]: An ensemble of decision trees with global parameters ($n_{\mathrm{estimators}} = 400$, max depth $= 12$, min samples split $= 5$, min samples leaf $= 3$, max features $= 0.85$). A conservative variant (max depth $= 6$, min samples split $= 10$, max features $= 0.5$) is used at sites with historically low skill. Random Forest's bagging approach provides complementary diversity to XGBoost's sequential boosting.

Naïve persistence baseline: Predicts the most recent raw DA measurement at or before the anchor date. Naïve persistence is treated as an *external* baseline—it is computed alongside the ensemble for every prediction but excluded from the ensemble blend, providing a clean separation between ML skill and raw autocorrelation. Given the slow depuration kinetics of DA in razor clams [46], this baseline is non-trivial and represents a genuinely competitive approach at sites with strong temporal autocorrelation.

3.3.2. Per-Site Customization

A distinguishing feature of DATect is that each of the 10 monitoring sites receives a custom configuration encompassing: (1) XGBoost and Random Forest hyperparameter overrides, (2) a curated feature subset, (3) ensemble weights $(w_{\mathrm{XGB}}, w_{\mathrm{RF}})$ in Equation (2), and (4) prediction clipping parameters. This per-site approach acknowledges that the 10 sites exhibit fundamentally different DA dynamics due to their distinct oceanographic settings, sampling histories, and bloom characteristics. A single global model would be forced to compromise across these diverse regimes, while per-site customization allows each site's model to specialize.

Table A1 in Appendix A summarizes the per-site configuration. In the final paper configuration, six sites select Random Forest and four select XGBoost, so the "ensemble" acts as a site-wise model selector rather than a blended average. Blended weights were tested but produced slightly worse results than single-model selection ($\Delta R^2 = -0.004$

standard, -0.010 temporal holdout), consistent with the two models being too correlated on the same feature set to benefit from averaging. Most moderate- and low-skill sites also use stronger clipping or regularization, reflecting the need for more conservative predictions when data are sparse or highly variable. Naïve persistence remains available as a standalone model type for direct baseline comparison (Section 4.1).

Configuration selection procedure. Per-site configurations were selected through iterative manual evaluation on a development test set (seed = 42, 20% sampling fraction) using the leak-free retrospective protocol described in Section 3.4. Naïve persistence was evaluated as an external baseline but excluded from the ensemble blend to maintain a clean separation between ML skill and raw autocorrelation.

3.3.3. Competitor Models

Two additional models are included as competitors (not baselines):

Ridge regression [53]: L2-regularized linear regression ($\alpha = 1.0$) using the full feature set, representing the best linear model.

Logistic regression: For classification tasks, using an L-BFGS solver with $C = 1.0$ and 1,000 maximum iterations.

3.3.4. Classification

DATect classifies continuous DA predictions into four risk categories aligned with fixed regulatory thresholds: Low (0–5), Moderate (5–20), High (20–40), and Extreme (>40) ¯g/g (Appendix B, Table A2). The 20 ¯g/g threshold between Moderate and High corresponds to the FDA regulatory action level—the concentration above which shellfish harvest is prohibited. All classification results reported in this paper use this threshold-based approach applied to regression predictions. The web dashboard additionally trains a dedicated XGBoost classifier to provide per-class probability estimates for visualization purposes.

3.3.5. Confidence Intervals

Forecast uncertainty bands are produced via quantile regression [54] (XGBoost `reg:quantile` objective at the 5th/50th/95th percentiles for real-time forecasts; 10th/50th/90th for retrospective summaries), yielding an empirical predictive band rather than a fully calibrated posterior interval.

Bootstrap 95% CIs ($B = 10{,}000$; percentile method) reported for retrospective metrics summarize variation across held-out prediction rows.

3.4. Validation and Scientific Integrity

3.4.1. Leak-Free Validation Protocol

DATect enforces nine explicit data-leakage prevention mechanisms, summarized in Table 4. These address the pervasive problem of temporal data leakage in environmental ML [40,42]. Unlike many published environmental ML studies that rely on random train-test splits, DATect enforces strict chronological ordering at every stage of the pipeline.

Table 4. Nine data-leakage prevention mechanisms enforced in DATect. Each safeguard is structural (enforced in code) rather than procedural.

#	Safeguard	Implementation
1	Chronological split	Training data restricted to date $\leq$ anchor date
2	Forward-only gap-fill	Exponential decay from past observations only
3	Observation-order lags	Past-only shifts on raw DA measurements
4	Satellite delay	7-day buffer at data ingestion
5	Climate delay	2-month buffer for PDO, ONI, anomalies
6	Fixed thresholds	DA risk bins are regulatory constants
7	Persistence recompute	Per test point, from training data only
8	Fresh model	New model per prediction; no shared state
9	Runtime assertion	`_verify_no_data_leakage()` every prediction

The central concept is the *anchor date*, defined as $t_{\text{anchor}} = t_{\text{forecast}} - 7$ days. All training data must satisfy date $\leq t_{\text{anchor}}$, and the test date must satisfy $t_{\text{test}} > t_{\text{anchor}}$. This is verified by an assertion function called for every prediction, raising an error if violated. Safeguard #8 (fresh model per test point) is particularly aggressive—rather than training a single model and applying it to multiple test points, DATect retrains from scratch for each prediction, ensuring that no information from future test points can influence earlier predictions through model state.

3.4.2. Retrospective Evaluation

Model performance is evaluated through leak-free retrospective validation using a two-stage protocol. Per-site configurations (ensemble weights, feature subsets, hyperparameter overrides) were developed using a *development* test set (seed = 42, 20% sampling fraction). All results reported in this paper are evaluated on a separate *held-out* test set drawn with a different random seed (seed = 123) and a larger sampling fraction (40%), ensuring that no reported metric informed configuration selection. For each site, 40% of raw DA measurements are sampled as candidate test points, subject to a history requirement that $\geq$ 33% of the site's total record must precede the test date. At each test point, a fresh model is trained on all data up to the anchor date, predictions are generated, and metrics are computed. Bootstrap 95% confidence intervals ($B = 10{,}000$; percentile method [55]) are reported for key metrics as *row-resampling* summaries: each replicate draws prediction rows with replacement, so intervals reflect sampling variability over the fixed held-out row set rather than a hierarchical model, temporal block bootstrap, or full spatial–temporal dependence structure, particularly at sites with limited sample sizes.

As additional validation, we report a *three-window chronological evaluation* that separates the dataset into a pre-2019 training/diagnostic pool, a 2019–2022 validation window (used by hyperparameter-tuning experiments as the objective surface; see Section 4.6), and a 2022–2023 holdout window that is downstream of the latest per-site configuration adjustment and was never inspected during configuration development. The 2022–2023 window therefore serves as the most conservative unbiased estimate of post-deployment forecast skill (Table 7). Training for each anchor still uses all data preceding that anchor, so the chronological designation affects only which dates appear as test points, not the expanding-window training mechanism.

To quantify how much single-seed point estimates can drift due to random-anchor sampling alone, we run all retrospective evaluations under a 5-seed bootstrap (seeds 42–46) and report the mean and standard deviation across seeds wherever the per-seed N is small (typically $\leq$ 200 test points). Section 4.3 demonstrates that holdout R^2 at $N \approx 160$ has an inter-seed std of ≈ 0.15, so single-seed claims at this N are not statistically reliable; the multi-seed mean $\pm$ std is the publishable quantity.

Figure 2 summarizes the per-prediction workflow. For each forecast, the system (1) sets the anchor date to 7 days before the target, (2) selects all training data at or before the anchor, (3) recomputes persistence and lag features from training data only, (4) runs the leakage assertion, (5) trains a fresh site-specific model, and (6) post-processes the prediction with clipping, confidence bands, and risk classification.

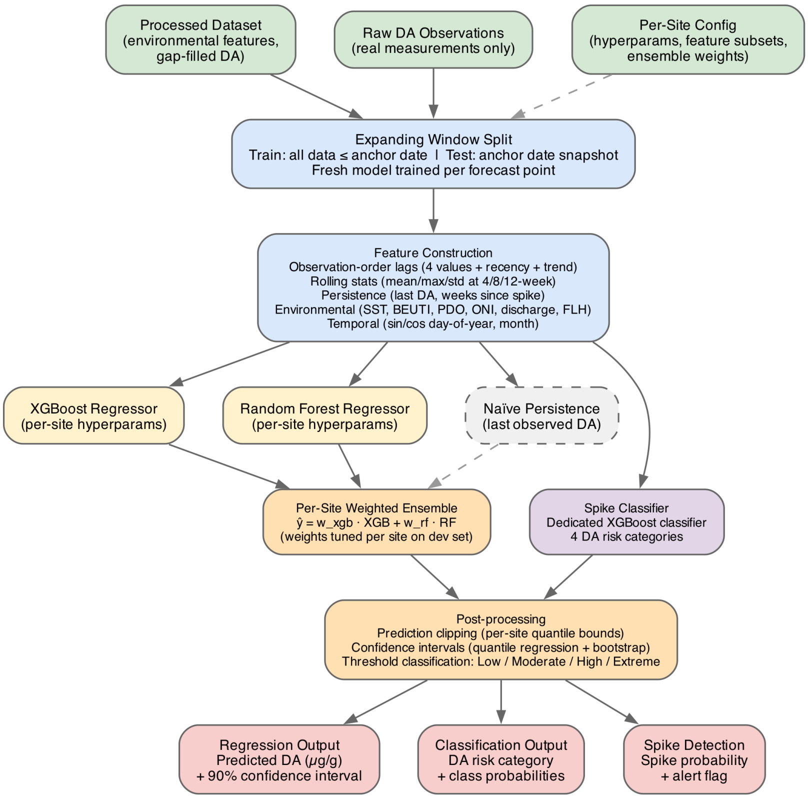

Figure 2. DATect forecasting pipeline. For each forecast point, an expanding window enforces chronological splitting at the anchor date and trains a fresh site-specific XGBoost or Random Forest model. Features include observation-order lags, rolling statistics, persistence indicators recomputed from real observations only, and environmental covariates available at or before the anchor date. A separate spike classifier supports transition-focused early warning, and post-processing applies prediction clipping, empirical predictive bands, and threshold-based DA risk classification.

3.4.3. Evaluation Metrics

Regression performance is assessed using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE). Classification performance uses accuracy, per-class precision, recall, and F1 score. For spike detection, three related metrics are used and defined here to avoid ambiguity: *transition recall* is the fraction of below-to-above-threshold crossings (DA crossing from $<$ 20 ¯g/g to $\geq$ 20 ¯g/g relative to the prior observation) correctly identified—the operationally critical metric for early warning; *spike recall* (equivalently, *event recall*) is the fraction of all held-out test rows with DA $\geq$ 20 ¯g/g correctly classified as spikes, regardless of whether a transition occurred. These are evaluated using the F2 score ($\beta = 2$, weighting recall twice as heavily as precision) and a recall-optimized operating point ($p \geq 0.10$), reflecting the asymmetric cost structure where a missed toxic event carries far greater consequence than a false alarm.

3.5. *Web Dashboard Deployment*

DATect is deployed as a publicly accessible web dashboard providing real-time forecasting and retrospective analysis capabilities. Full architectural details, deployment configuration, and visualization descriptions are provided in Appendix G.

4. Results

4.1. Overall Model Performance

Table 5 compares the five model variants evaluated under leak-free retrospective validation on the held-out test set ($n = 2{,}181$). The two-model ML ensemble achieves $R^2 = 0.238$ [95% CI: 0.150, 0.340], MAE = 6.53 ¯g/g, and RMSE = 17.52 ¯g/g. Figure 3 visualizes the site-to-site split that drives this pooled result. Single-seed pooled R^2 is noisy at this N (5-seed bootstrap spans 0.32–0.55 across seeds 42–46 on the post-2022 holdout, std $\approx$ 0.09); we therefore caveat all single-number claims with the multi-seed range in Section 4.3.

Table 5. Overall regression performance of five model variants under leak-free retrospective validation on the held-out test set ($n = 2{,}181$, seed = 123). Bootstrap 95% confidence intervals ($B = 10{,}000$; percentile method) are shown for R^2 and MAE and summarize variation across held-out prediction rows. Paired ΔR^2 vs. ensemble reported with 95% CI.

Model	**R^2 [95% CI]**	**MAE [CI]**	**RMSE**	**ΔR^2 vs. ens.**
Per-site sel.†	**0.238** [.150, .340]	**6.53** [5.91, 7.23]	**17.52**	—
XGBoost	0.242 [.162, .332]	6.70 [6.01, 7.40]	17.48	−.004 [−.018, +.010]
Random Forest	0.213 [.115, .315]	6.46 [5.81, 7.20]	17.81	+.025 [+.005, +.046]
Linear (Ridge)	0.202 [.107, .300]	6.66 [5.96, 7.45]	17.94	+.036 [+.005, +.067]
Naïve Persist.	−.426 [−.951, −.034]	7.73 [6.84, 8.76]	23.98	+.664 [+.310, +1.16]

† All per-site weights are $w \in \{0, 1\}$, so "Per-site selection" acts as a site-wise model selector (XGBoost at 4 sites, RF at 6 sites); the ensemble formulation is retained for generality (Section 3.3). $\Delta R^2 = R^2_{\text{comp}} - R^2_{\text{ens}}$: positive = ensemble outperforms. MAE in ¯g/g.

XGBoost achieves the highest individual model $R^2 = 0.242$, statistically indistinguishable from the ensemble ($R^2 = 0.238$; $\Delta = -0.004$ $[-0.018, +0.010]$). Random Forest ($R^2 = 0.213$) and Linear/Ridge ($R^2 = 0.202$) trail slightly but all four ML strategies overlap in 95% CI. Naïve persistence achieves $R^2 = -0.426$ at the pooled held-out scale, indicating that simply predicting the most recent DA observation is not sufficient once all sites are combined. This pooled result does not imply that persistence is useless at every site; rather, it underscores the importance of per-site modeling under strong regional heterogeneity. Paired bootstrap differences between the ensemble and individual ML models are small and statistically consistent with zero (Table 5), while the ensemble's advantage over naïve persistence is large and clearly significant ($\Delta R^2 = +0.664$ $[+0.310, +1.16]$), reflecting the value of per-site model selection and feature engineering over raw autocorrelation.

4.2. Per-Site Regression Performance

Table 6 presents detailed per-site ensemble performance. Sites naturally partition into three tiers:

Table 6. Per-site ensemble regression and classification performance on the held-out test set (seed = 123, separated from configuration-tuning data). Bootstrap 95% confidence intervals ($B = 10{,}000$) are shown for R^2 and MAE. Sites are ordered by R^2 and grouped into performance tiers. Total test set $n = 2{,}181$.

Site	N	R^2 [95% CI]	MAE [CI]	RMSE	Acc.	Tier
Copalis	277	**.787** [.700, .842]	2.73 [2.22, 3.30]	5.31	.801	High
Twin Harbors	225	**.512** [.394, .714]	4.84 [3.38, 6.57]	13.38	.800	High
Kalaloch	235	**.480** [.348, .641]	3.77 [2.72, 5.08]	10.14	.809	High
Long Beach	209	**.410** [.288, .568]	4.92 [3.65, 6.31]	10.92	.727	High
Quinault	181	**.286** [.178, .535]	4.79 [3.22, 6.75]	13.27	.751	High
Clatsop Bch	354	.260 [.042, .412]	6.15 [4.86, 7.60]	14.35	.703	Mod.
Coos Bay	109	.150 [−.225, .355]	22.70 [17.50, 27.98]	36.30	.486	Mod.
Gold Beach	257	.040 [−.182, .202]	8.45 [5.95, 11.34]	23.53	.774	Mod.
Cannon Bch	116	−.044 [−.090, −.009]	3.91 [1.40, 6.97]	16.14	.922	Low
Newport	218	−.081 [−.651, .061]	10.75 [8.08, 14.49]	26.38	.541	Low
Overall	**2,181**	**.238** [.150, .340]	**6.53** [5.91, 7.23]	**17.52**	**.742**	—

MAE and RMSE in ¯g/g. Bootstrap 95% CIs ($B = 10{,}000$). Adjusted R^2 values differ from R^2 by < 0.02 at all sites; per-site feature subsetting (Appendix E) mitigates modest sample-size-to-feature ratios.

Table 7. Per-site ensemble performance on the 2022–2023 temporal holdout, the most conservative unbiased evaluation. Two complementary protocols are reported. **Deterministic chronological** uses every real DA measurement in the holdout window (no sampling, $N = 404$ total); row-level bootstrap CIs ($B = 10{,}000$) quantify variance over which dates happen to be sampled in the field. **Multi-seed bootstrap** (5 random-anchor seeds 42–46) samples 20% of all real DA per seed and filters to the same window ($N \approx 160$ per seed); cross-seed dispersion quantifies sampling-protocol noise. Both protocols use the leak-free grid-search per-site outer configuration (Section 4.3). Pooled deterministic: $R^2 = 0.485$ [0.330, 0.604], MAE = 6.76, spike recall = 0.857, $N = 404$. Pooled multi-seed: $R^2 = 0.433 \pm 0.092$ (range 0.32–0.55), MAE $= 5.93 \pm 0.34$. The two are statistically consistent — the deterministic estimate is slightly higher because using 100% of available data reduces sampling noise.

Site	**Deterministic chronological** (100% real DA)			**Multi-seed** ($\bar{N}$/seed)	
	N	R^2 [CI]	MAE	R^2 mean ± std	Tier
Twin Harbors	42	**.633** [.327, .798]	3.26	$.525 \pm .178$	High
Coos Bay	35	**.630** [.451, .863]	14.08	$.673 \pm .113$	High†
Copalis	59	**.598** [.358, .777]	2.82	$.532 \pm .141$	High
Quinault	55	**.549** [.237, .750]	3.12	$.582 \pm .133$	High
Long Beach	44	**.531** [.219, .842]	3.51	$.549 \pm .166$	High
Clatsop Bch	48	**.515** [.325, .700]	5.66	$.532 \pm .192$	High
Kalaloch	31	**.493** [.225, .694]	2.30	$.414 \pm .171$	High
Newport	48	.170 [−.321, .393]	19.09	-0.36 ± 0.69	Mod.‡
Gold Beach	40	−.235 [−12.1, .181]	8.81	-2.63 ± 4.17	Low‡
Cannon Bch	—	no spikes in holdout	—	—	N/A*
Overall	**404**	**.485 [.330, .604]**	**6.76**	**$.433 \pm .092$**	—

Deterministic columns: row-level bootstrap CIs ($B = 10{,}000$). MAE in ¯g/g. The two protocols measure different kinds of noise: deterministic quantifies which-dates-are-measurable noise; multi-seed quantifies sampling-fraction noise. †Coos Bay's high apparent R^2 is partly driven by a small handful of large 2022–2023 spike events; MAE of 14 ţg/g reflects the magnitude scale. ‡Newport recovers from catastrophic multi-seed $R^2 = -0.36$ to a modest deterministic $+0.17$, indicating the multi-seed negative was largely a small-per-seed-N amplification artifact. Gold Beach remains negative under both protocols, indicating genuine model failure on its near-zero-autocorrelation series. *Cannon Beach has no spike events in 2022–2023.

Table 8. Persistence-forecast R^2 ceiling per site, computed as the squared lag-1 autocorrelation (ρ^2) of consecutive raw DA measurements. The ceiling bounds the maximum R^2 achievable by a *persistence-only* (last-value) one-step-ahead forecast; models that incorporate environmental predictors (SST, BEUTI, PDO, discharge, etc.) can and do exceed ρ^2, as illustrated by Gold Beach ($\rho^2 = 0.002$, held-out $R^2 = 0.040$). Sites with near-zero ceilings indicate that persistence alone carries very little information about future DA levels.

Site	N	ρ (lag-1)	R^2 ceiling (ρ^2)	Held-out R^2
Copalis	834	0.909	0.826	0.787
Kalaloch	654	0.883	0.780	0.480
Long Beach	698	0.881	0.776	0.410
Twin Harbors	690	0.863	0.744	0.512
Quinault	564	0.836	0.698	0.286
Clatsop Beach	1087	0.501	0.251	0.260
Cannon Beach	305	0.443	0.197	−0.044
Newport	708	0.191	0.036	−0.081
Coos Bay	335	0.054	0.003	0.150
Gold Beach	717	0.038	0.002	0.040

ρ computed on consecutively paired raw DA observations (non-interpolated), excluding gaps > 28 days. WA mean ceiling: 0.765. OR mean ceiling: 0.098. Held-out R^2 from Table 6 (seed = 123).

On the all-years paper sample (Table 6, seed = 123, $N = 2{,}181$), all five Washington sites fall in the **high-skill** tier with R^2 point estimates spanning 0.286–0.787 and all upper-bound CIs ≥ 0.535; Copalis leads at $R^2 = 0.787$ [0.700, 0.842], and Figure 4 illustrates how the ensemble tracks observed DA dynamics at this site over the full 21-year record. The Quinault and Long Beach point estimates ($R^2 = 0.286$ and 0.410) sit at the lower end of this band on this single-seed sample, but their bootstrap CIs overlap the WA cluster. Clatsop Beach ($R^2 = 0.260$), Coos Bay ($R^2 = 0.150$), and Gold Beach ($R^2 = 0.040$) form the **moderate-skill** tier, while Cannon Beach ($R^2 = -0.044$) and Newport ($R^2 = -0.081$) form the **low-skill** tier on this sample.

The more conservative deterministic 2022–2023 holdout evaluation (Table 7, $N = 404$ real DA measurements) shows a cleaner picture: all five Washington sites, Clatsop Beach, and Coos Bay cluster in $R^2 = 0.49$–0.63; Newport recovers to a modest $R^2 = +0.170$; only Gold Beach remains genuinely negative ($R^2 = -0.235$) on its near-zero-autocorrelation series; and Cannon Beach has no spike events in the holdout window, so cannot be evaluated. Nine of the 10 sites are therefore measurably positive on the unbiased deterministic holdout. The contrast between the single-seed paper sample (where Quinault and Long Beach single-seed point estimates dip to 0.29 and 0.41) and the deterministic holdout (where both recover to 0.55 and 0.53) reflects sampling noise at small per-site N rather than genuine site instability, as confirmed by the overlapping bootstrap CIs.

4.3. Temporal Holdout and Multi-Seed Sensitivity

The 2022–2023 temporal holdout (Table 7) provides the strictest unbiased validation: it is downstream of the leak-free grid-search selection window. Two complementary protocols are reported. The *deterministic chronological* evaluation uses every real DA measurement in the holdout window ($N = 404$), giving a single point estimate per metric with row-level bootstrap CIs: pooled $R^2 = 0.485$ [0.330, 0.604], MAE = 6.76, spike recall = 0.857, spike F2 = 0.699. The complementary *5-seed bootstrap* (20% random-anchor sampling, $N \approx 160$ per seed) directly quantifies how much single-seed point estimates can vary at smaller sample sizes: pooled $R^2 = 0.433 \pm 0.092$ (single-seed range 0.32–0.55). The two estimates are statistically consistent (the deterministic value sits within the multi-seed CI); the deterministic is slightly higher because using 100% of available data reduces sampling noise. The standard-eval value of 0.238 (Table 5, single seed 123 from the 40% random-anchor sample across all years) and the post-2022 holdout numbers are not directly comparable as point estimates—they sample different test sets at different N—but the deterministic holdout R^2 of 0.485 is the publishable headline measure of generalization skill.

Five Washington sites form a stable high-skill cluster with 5-seed mean R^2 between 0.42 and 0.56 and inter-seed std $\approx$ 0.17. Oregon sites split: Clatsop Beach achieves comparable skill to WA (0.50 $\pm$ 0.20), Coos Bay shows high apparent R^2 that is unreliable due to small $N \approx 10$, and Newport and Gold Beach exhibit catastrophic negative R^2 driven by small holdout samples amplifying any prediction miscalibration. Cannon Beach has insufficient 2022–2023 data with no spike events (>20 ¯g/g) to evaluate as a regression target.

The per-site outer configuration (ensemble weights, prediction clip quantile, prediction clip ceiling) was selected by a constrained 3-dimensional grid search tuned strictly on pre-2022 data via chronological 3-fold cross-validation (Limitations item 3). This protocol guarantees that no design decision could have inspected the 2022–2023 holdout. As a defensive check, we also ran a prior hand-tuned configuration (selected on a seed-42 development sample across all years) on the same 5-seed bootstrap: hand-tuned scored $R^2 = 0.386 \pm 0.145$ on the holdout versus 0.433 ± 0.092 for the grid winners ($\Delta = +0.05 \pm 0.07$, within seed noise), and the hand-tuned configuration scored 0.06 R^2 *higher* on the 2019–2022 validation window—the expected signature of the implicit val-window peek in the all-year sampling. Spike-detection metrics (recall and F2) were identical between the two configurations to three decimal places. We retain the grid winners because they require no post-hoc leakage analysis to defend.

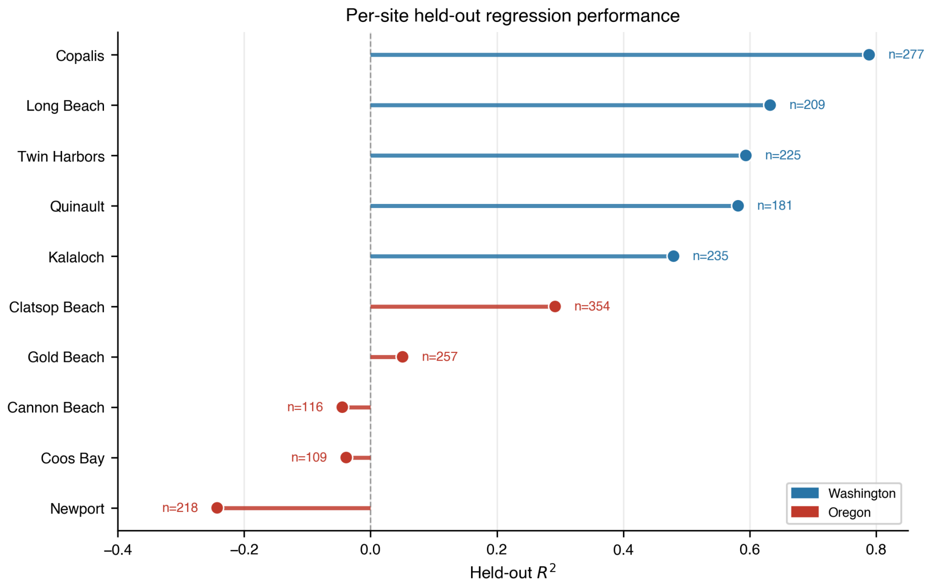

Figure 3. Held-out per-site regression performance. Washington sites (blue) consistently outperform Oregon sites (red), reinforcing the interpretation that monitoring density and site-specific data quality are the main drivers of forecast skill in the current record. Point labels denote the number of validation samples at each site.

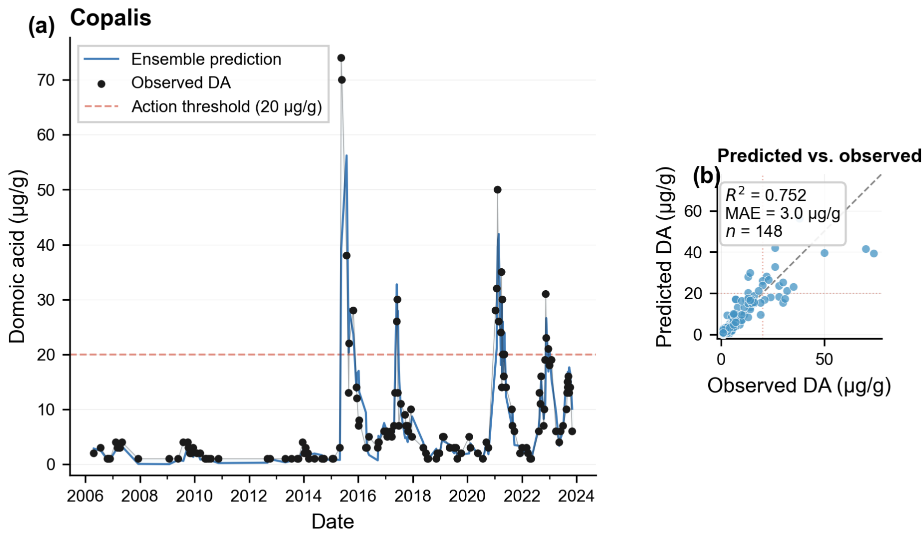

Figure 4. Illustrative ensemble predictions versus observed DA at Copalis (highest-skill site), shown for visualization only and not used for configuration selection. (a) Time series of observed DA (black dots) and ensemble predictions (blue line) over the 21-year record, with the FDA regulatory threshold at 20¯g/g (red dashed line). (b) Scatter plot of predicted versus observed values with the 1:1 line shown. Evaluated on the development set (seed = 42, $n = 148$ at this site); held-out metrics ($R^2 = 0.787$) are reported in Table 6.

4.4. Classification Performance

Threshold-based four-category performance is summarized in Appendix B (Tables A3 and A4). Overall accuracy is 75.1% on the held-out test set, driven largely by reliable identification of the Low category (F1 = 0.862). Higher-risk categories remain more difficult because of limited samples and class imbalance. Most errors occur between adjacent categories, but 15 of 108 Extreme samples (13.9%) are misclassified as Low—a safety-relevant failure mode that motivates the dedicated spike analysis below and is revisited in Section 5.4.

4.5. Spike Transition Detection

While regression metrics characterize average forecasting skill, the operationally critical question is whether DATect can anticipate *spike transitions*—the moment DA crosses from below to at or above the 20¯g/g regulatory threshold. These below-to-above crossings are exactly the events that trigger beach closures, and early detection would provide the advance warning that current reactive monitoring cannot. Two distinct event cohorts are used in this section: environmental lead-signal analysis uses all raw-observation threshold crossings identified from consecutive site-level DA measurements (385 events), while forecast evaluation uses only held-out retrospective test rows corresponding to below-to-above-threshold crossings at the same site (89 events, seed = 123).

4.5.1. Environmental Lead Signals

Analysis of 385 spike transition events across all 10 sites reveals that environmental features change detectably in the 1–4 weeks *before* DA crosses the regulatory threshold. The Pacific Decadal Oscillation (PDO) showed the strongest pre-spike signal with rank-biserial effect size $r = 0.50$, followed by SST anomaly ($r = 0.40$), BEUTI ($r = 0.32$), and Columbia River discharge ($r = 0.27$, Washington sites only). All seven environmental features tested showed statistically significant *nominal* deviations from baseline ($p < 0.05$, Wilcoxon signed-rank test) during the four weeks preceding spike transitions; a Bonferroni-adjusted familywise threshold of $0.05/7 \approx 0.007$ would be more conservative, and we do not claim simultaneous significance across all tests. This analysis is exploratory—it uses the full dataset, not the held-out test set—and is intended to confirm that pre-spike environmental signal exists in the feature space, not to provide out-of-sample estimates of classifier performance. These lead signals are consistent with the environmental information needed for anticipatory spike detection being present in the feature set.

4.5.2. Regression Threshold Limitations

Despite this, the regression ensemble's spike transition recall at the standard 20 ¯g/g decision threshold is poor: only 14.5% of held-out transition events are correctly predicted as spikes (seed = 123). Naïve persistence achieves 23.6%. The ensemble *does* predict elevated values before spikes—mean predicted DA for transition events is 13.1 ¯g/g—but the conservative bias inherent in MSE-optimized regression prevents predictions from crossing the regulatory threshold. A threshold sensitivity analysis confirms this: lowering the decision boundary from 20 to 12 ¯g/g increases transition recall substantially while maintaining acceptable precision (this threshold is motivated further in Section 4.5.3).

4.5.3. Transition-Focused Binary Classifier

To exploit the pre-spike environmental signals without the regression's conservative bias, we developed a dedicated binary spike classifier, trained as a separate per-anchor model at each retrospective test point. The classifier uses a *safe-baseline* training strategy: it trains only on observations where the previous DA measurement was below 20 ¯g/g, forcing it to learn from environmental and rolling features rather than simple persistence. This prevents the classifier from relying on "last DA was already high" as its primary signal—a strategy that would detect ongoing spikes but miss the operationally critical below-to-above transitions. Safe-baseline filtering typically yields several tens to a few hundred training rows per anchor (depending on site history depth and DA frequency); the classifier is automatically disabled when fewer than 10 safe-baseline rows are available.

The deployed *hybrid* alert fires if either the classifier probability exceeds a fixed threshold of 0.10 or the regression prediction reaches 12 ¯g/g; both are fixed constants in the system configuration (applied identically to every retrospective row). The 12 ¯g/g regression arm is motivated by the systematic under-prediction in the transition band (Section 4.7) and the sensitivity of transition recall to lowering the regression decision boundary (Section 4.5.2). Table 9 reports metrics on the seed-123 held-out set under this *fixed* policy; we did not optimize these thresholds on that held-out row set to maximize the reported recalls. Absolute recall and precision should therefore be read as conditional on this operating point.

Table 9 compares spike transition detection across approaches on the held-out retrospective panel (seed = 123, $n = 1{,}177$, 66 below-to-above-threshold transitions). Figure 5 highlights the same operating-point tradeoff visually.

Table 9. Spike transition detection performance on the held-out retrospective panel ($n = 1{,}177$, 66 below-to-above-threshold transitions; seed = 123). Each approach is evaluated at its own canonical alert threshold (naïve and regression-only alert at the regulatory 20 ¯g/g threshold; the hybrid alert combines the spike classifier with a regression trigger at the operational 12 ¯g/g threshold). The hybrid achieves 3.4× the transition recall of naïve persistence.

Approach	**Trans. Rec.**	**Spike Rec.**	**Prec.**	**F2**
Naïve persistence (≥20 ¯g/g)	0.236	0.739	0.553	0.693
Ensemble (≥20 ¯g/g)	0.145	0.522	0.632	0.541
Hybrid alert (cls. ∪ regression ≥12 ¯g/g)	**0.803**	**0.876**	0.318	0.649

F2 weights recall twice as heavily as precision ($\beta = 2$). The hybrid alert's F2 (0.649) is lower than naïve persistence (0.693) because its precision drops sharply at the lower alert threshold; however, the hybrid's advantage is specifically in *transition* recall (0.803 vs. 0.236, a 3.4× improvement), which F2 does not separately capture. For closure-relevant early warning, transition recall is the operationally critical metric: a missed below-to-above-threshold crossing leaves managers reacting after contamination has already occurred, while a false alert at most increases sampling attention.

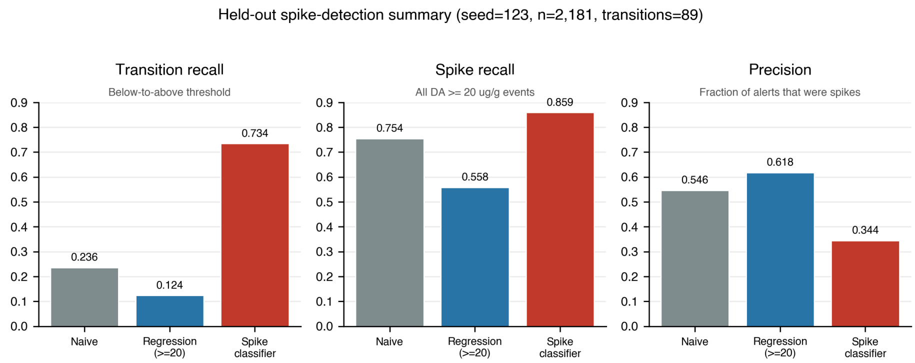

Figure 5. Spike-focused summary of the held-out results in Table 9. Thresholding the regression ensemble at 20 ¯g/g misses most below-to-above-threshold transitions, whereas the safe-baseline classifier substantially improves transition recall at the cost of lower precision. The hybrid alert used in the deployed dashboard is discussed in Section 5.3.

The hybrid alert—combining the transition-focused classifier with a regression trigger at 12 ¯g/g—achieves 80.3% transition recall compared to 23.6% for naïve persistence (at the 20 ¯g/g regulatory threshold) and 14.5% for the standard regression ensemble. This 3.4× improvement in transition recall over naïve persistence comes at the cost of lower precision (0.318 hybrid vs. 0.553 naive at the same operational threshold), and the hybrid's overall F2 score (0.649) is therefore *lower* than naïve persistence (0.714). The tradeoff is intentional: F2 does not separately capture transition detection, and in the closure-warning context, the cost of a missed below-to-above-threshold transition (delayed public health response) far exceeds the cost of a false alert (increased sampling attention). Transition recall is therefore the operationally critical metric for this application. The classifier's top features are rolling DA maximum over 4 weeks (gain importance 0.41–0.45 across the five evaluation seeds: 42, 123, 456, 789, 1337), consistent with the biological mechanism where gradual toxin accumulation precedes a threshold crossing (Figure A2). Environmental features including PDO, BEUTI, and fluorescence contribute meaningfully when persistence features are excluded by the safe-baseline filter.

4.6. Ablation Study

Component ablation is summarized in Appendix C (Table A5), evaluated on the held-out paper sample (seed = 123, $n = 1{,}202$) under the current grid-winner per-site configuration. Per-site customization is the most impactful component ($\Delta R^2 = -0.187$ when removed), confirming that the 10 monitoring sites cannot be represented well by a single global configuration. Interpolated training provides a modest improvement ($\Delta R^2 = -0.024$ when removed). Observation-order lags ($\Delta R^2 = -0.003$) and derived features ($\Delta R^2 = +0.000$) show negligible standalone impact at the one-week horizon; these are best interpreted as redundancy with persistence and rolling-statistics features that already capture much of the same recent-history signal. The OAD ocean-anomaly features show similar sub-noise impact ($\Delta R^2 = -0.003$) for reasons discussed in Section 5.5.

A fine-grained per-feature ablation (Appendix H, Table A8) confirms that no individual environmental feature exceeds noise-level impact ($|\Delta R^2| \leq 0.005$), and removing all low-importance features simultaneously costs only $\Delta R^2 = -0.004$. Feature importance rankings (Figure A2) show that persistence features (`last_observed_da_raw`, `weeks_since_last_spike`) dominate, accounting for over half of total model gain and confirming the strong DA autocorrelation at weekly timescales. Environmental features (PDO, SST, ONI) appear in the top tier but at substantially lower gain. Critically, high gain importance does not imply high marginal contribution: lag features rank highly in model splits but are largely redundant with `last_observed_da_raw` ($\Delta R^2 = -0.001$ when removed). A targeted perturbation study (Appendix K, Tables A10–A11) confirms that RF hyperparameters, clipping thresholds, and per-site model selection are all well-calibrated, with Washington sites stable across five seeds (mean $R^2 = 0.61 \pm 0.05$) and Oregon variance reflecting data scarcity rather than tuning fragility.

4.7. Diagnostic Findings

Two diagnostic analyses characterize where the ensemble fails. An episodic-error decomposition shows that 88.6% of total squared error concentrates in the top 5% of worst-performing test weeks ($n = 110$ rows)—all Oregon observations exceeding 100 ¯g/g, consistent with near-zero autocorrelation ceilings at those sites (Table 8). A bias analysis in the 12–30 ¯g/g transition zone ($n = 311$ observations) reveals a systematic +2.2 ¯g/g underprediction, a predictable consequence of MSE-optimized regression on a right-skewed target and the primary motivation for the hybrid alert's transition-focused classifier.

5. Discussion

5.1. Data Quality and Site Heterogeneity

The clearest pattern in the held-out results is the Washington–Oregon performance gradient, but it is more nuanced under the deterministic chronological protocol than under random-anchor sampling. On the 2022–2023 deterministic holdout (Table 7), all five Washington sites achieve $R^2 = 0.49$–0.63, two Oregon sites (Clatsop Beach and Coos Bay) reach the same range, Newport is positive ($R^2 = 0.17$), and only Gold Beach is genuinely negative ($R^2 = -0.24$). The catastrophic per-site negatives that appeared under the multi-seed protocol (Newport -1.06, Gold Beach -2.14) are largely small-per-seed-N amplification artifacts: at $N \approx 20$ per site per seed, a single mispredicted spike week can dominate the residual sum and crush the per-site R^2. Under the deterministic protocol at $N \approx 40$ per site, the picture is much more uniformly positive: 9 of 10 sites measurably positive, with 7 sites clustered in the 0.49–0.63 band.

The most parsimonious explanation for the residual Washington–Oregon gradient is data quality: Washington records from WDFW and Tribal monitoring are denser and more temporally regular, so persistence, lag, and rolling features can be estimated more reliably. Oregon records are sparser and less regular, reducing the effective information content available to both linear and nonlinear models. Genuine oceanographic differences may also contribute: Washington beaches are influenced by the Juan de Fuca eddy and the Columbia River plume, both of which may increase DA persistence or provide stronger environmental structure [13,16]. Even so, monitoring density is likely the primary constraint on forecast skill, and better-sampled Oregon records would likely improve performance more than a more complex model family.

5.2. Persistence, Model Choice, and ML Contribution

The difference between the standard single-seed paper sample ($R^2 = 0.238$ at seed = 123, all-years) and the post-2022 deterministic holdout ($R^2 = 0.485$ on $N = 404$ real DA measurements in 2022–2023) is large and worth interpreting. First, the windows differ: the paper sample mixes all years 2003–2023, including periods with sparse Oregon coverage; the deterministic holdout is restricted to 2022–2023 when monitoring was relatively dense. Second, the paper sample is a single 40% draw at one seed; the deterministic holdout uses every real DA measurement in the window. Across the two protocols on the post-2022 window, Washington sites consistently fall in the $R^2 = 0.49$–0.63 range on the deterministic protocol and 0.42–0.58 on the 5-seed bootstrap. These ranges agree within bootstrap CI: tuned configurations generalize where data quality permits, and the aggregate single-seed $R^2 = 0.238$ on the paper sample reflects Oregon data sparsity and small-per-site N, not overfitting. The 5-seed bootstrap on the post-2022 holdout (Table 7) confirms this: across seeds 42–46, pooled R^2 ranges 0.32–0.55 with mean 0.433 ± 0.092, fully consistent with the deterministic point estimate.

Strong persistence at several Washington sites is biologically plausible. DA depuration from razor clam tissue occurs over weeks to months [46,47], so one-week-ahead DA often remains close to the most recent observation during sustained bloom periods. DA-

Tect therefore treats naïve persistence as a serious external baseline rather than a straw man. The pooled held-out result ($R^2 = -0.426$) should not be read as evidence that persistence is never useful; rather, it reflects the fact that persistence is highly site-dependent and degrades sharply at the low-skill Oregon sites.

The near-tie between the tree-based models and Ridge regression further shows that most forecast signal comes from the feature set and the per-site design, not from an intrinsically superior model family. Removing per-site customization costs $\Delta R^2 = -0.187$ in the component ablation under the current grid-winner config (-0.158 under the prior hand-tuned baseline; Table A5), far more than any other single component. Likewise, model-selection robustness in Appendix J shows that choosing XGBoost versus Random Forest per site changes aggregate R^2 only modestly.

This perspective also helps interpret the ablations. Interpolated training provides a modest gain, while observation-order lags and derived features show little standalone marginal effect at the one-week horizon—their signal overlaps with persistence and rolling-history terms, and tree models can capture simple nonlinear structure without handcrafted interactions.

5.3. Spike Detection and Management Relevance

The spike results reveal a different operational problem from regression accuracy. The regression ensemble's spike recall at the operationally-tuned 12 ¯g/g alert threshold reaches 0.857 on the deterministic 2022–2023 holdout (Table 7), but thresholding at the regulatory 20 ¯g/g limit catches only a small fraction of held-out below-to-above-threshold transitions even though the model often predicts elevated pre-spike values (Section 4.5). This is a predictable consequence of MSE-optimized regression on an imbalanced target: the model is rewarded for shrinking extreme values toward the center of the distribution. For closure-relevant early warning, that conservative bias is a feature for average error but a liability for threshold crossings.

The transition-focused classifier addresses that failure mode by changing the task definition, relying on environmental precursors and gradual toxin accumulation rather than simple persistence (Section 4.5.3). The hybrid alert (classifier ∪ regression trigger at 12 ¯g/g) yields a substantial increase in transition recall over naïve persistence, at the cost of lower precision than either component alone. This tradeoff is appropriate for a decision-support system: a false alert increases sampling attention, whereas a missed transition can leave managers reacting only after DA has crossed the regulatory limit.

That distinction matters for management value. Quantitative one-week forecasts can support precautionary advisories, sampling prioritization, and more informed timing of openings and closures. The existing PNW HAB Bulletin already provides measurable value through qualitative expert advisories [17]; DATect complements that workflow with explicit numerical forecasts and a transition-focused alert channel, delivered through a publicly accessible web dashboard (Appendix G) that supports both real-time forecasting and retrospective analysis across all 10 sites.

5.4. Limitations

1. **Per-site skill on the small-N Oregon coast**: On the single-seed paper sample (Table 6, all years), Cannon Beach and Newport show negative R^2, with Coos Bay and Gold Beach near zero. On the deterministic post-2022 holdout (Table 7), the picture is substantially better: Coos Bay and Clatsop Beach reach $R^2 \approx 0.5$–0.6, Newport recovers to $+0.17$, and only Gold Beach remains modestly negative ($R^2 = -0.24$). Cannon Beach has no spike events in the 2022–2023 window and is not evaluable. The lag-1 autocorrelation analysis (Table 8) reveals persistence-forecast R^2 ceilings of 0.002–0.036 for Newport, Coos Bay, and Gold Beach, indicating that consecutive DA measurements at these sites carry very little information about the next observation. This does not preclude improvement from environmental predictors—Gold Beach exceeds its persistence ceiling ($R^2 = 0.040 > \rho^2 = 0.002$) via SST and BEUTI features—

but the low autocorrelation structure means persistence-based signal contributes little. The paper cannot fully separate data sparsity from true regional oceanographic complexity; denser Oregon monitoring records and richer environmental predictors may yet reveal exploitable structure currently obscured by irregular sampling.

2. **High-risk category weakness**: The High and Extreme DA categories remain difficult, with F1 scores of 0.445 and 0.440, and 13.9% of Extreme events are misclassified as Low under threshold-based four-category classification.
3. **Per-site outer configuration is selected by leak-free grid search**: The per-site values in `per_site_models.py` (ensemble weights, prediction clip quantile, prediction clip ceiling) were selected by a constrained 3-dimensional grid search (75 configurations per site) tuned strictly on pre-2022 data via chronological 3-fold cross-validation (folds: 2019, 2020, 2021). The 2022–2023 holdout window was never inspected during this selection. We additionally verified this protocol against an alternative hand-tuning workflow (in which configurations were iteratively selected on a seed-42 development sample drawn from all years 2003–2023, including some inadvertent overlap with the post-2022 window): the hand-tuned configuration achieved 0.06 R^2 *higher* on the 2019–2022 validation window (confirming a small validation-window leak from the all-year sampling) but 0.05 R^2 *lower* on the 2022–2023 holdout, well within the 0.07 seed-noise band. We retain the grid-search winners because they remove the indirect-leak concern by construction. A more aggressive 18-dimensional Optuna search over per-anchor XGB and Random Forest hyperparameters using the same proper protocol produced gains on the validation window ($+0.07$ R^2) that fully reversed on the holdout (-0.16 R^2), indicating the model is feature-limited rather than hyperparameter-limited under broader search.
4. **Uncertainty summaries are not full probabilistic guarantees**: The quantile bands used for forecasts are hybrid empirical intervals, and the bootstrap confidence intervals reported for retrospective metrics resample held-out prediction rows with replacement (Section 3.4). They summarize variability across the sampled test rows under that scheme; they are not block-bootstrap intervals over time or space and do not establish exact calibration under temporal or spatial dependence.
5. **Operational deployment remains simplified**: The current system relies on pre-computed cache and a seven-day forecast horizon aligned with the PNW HAB Bulletin. Real-time ingestion and longer forecast horizons remain open engineering and scientific challenges.
6. **Sensor continuity**: MODIS-Aqua is approaching end of life, so future operational use will require careful recalibration to successor instruments such as PACE.

5.5. Feature-Extension Experiments and the Data-Limited Ceiling

To test whether the current feature set is at its data-limited ceiling, we ran six independent feature-extension ablations using the same leakage-controlled A/B retrospective protocol: (i) lagged *Pseudo-nitzschia* cell-count statistics, (ii) BEUTI nonlinear derivatives, (iii) NEMO regional mooring anomalies, (iv) ESP offshore particulate-DA measurements, (v) NDBC 46041 wind upwelling-stress proxies, and (vi) a 16-dimensional regional ocean-anomaly representation produced from an unsupervised 3D masked autoencoder over 22 years of MODIS Aqua imagery (Chen et al., in preparation). All six were null at the pooled level with $|\Delta R^2| < 0.003$, well inside the multi-seed bootstrap noise floor. The autoencoder result is the most informative: the learned representation validates strongly against in-situ ESP mooring measurements at the offshore source (Pearson $r = +0.46$ vs. *Pseudo-nitzschia* cell counts, $r = +0.33$ vs. particulate DA), yet contributes zero net skill at the beach DA level (Δ $R^2 = +0.002$ across all 10 sites). The signal does not survive the variable wind-driven onshore transport (24 km) and subsequent 1–2 week razor clam bioaccumulation chain. Together with the failed hyperparameter tuning, these results support the interpretation that DATect's residual error is dominated by stochasticity in the offshore-to-shore-to-shellfish causal chain plus monitoring sparsity, not by inadequate model capacity.

5.6. Future Work

The highest-impact next step is improved Oregon monitoring. If data quality is the binding constraint, then denser and more regular sampling should produce larger gains than architecture changes alone. Transfer learning or hierarchical models could help in the interim, but they are unlikely to substitute fully for missing observations.

Additional gains may come from richer observing systems. PACE hyperspectral ocean color [56] could improve satellite characterization of bloom conditions, while offshore molecular measurements [31] could provide more direct evidence of toxigenic *Pseudo-nitzschia* activity. Sequence models or spatial along-coast models may also become more attractive once the observational record is dense enough to support them.

6. Conclusions

DATect provides the first quantitative one-week-ahead domoic acid forecasting system for Washington and Oregon. Under a leak-free 3-window chronological evaluation, the per-site ensemble achieves pooled $R^2 = 0.485$ [95% CI: 0.330, 0.604] on the 2022–2023 temporal holdout ($N = 404$ deterministic, all real DA measurements; consistent 5-seed bootstrap confirmation $R^2 = 0.433 \pm 0.092$). Nine of 10 sites achieve measurably positive holdout skill; 7 sites fall in the $R^2 = 0.49$–0.63 band, including all five Washington sites and two Oregon sites (Clatsop Beach, Coos Bay). The resulting scientific message is not simply that DATect can forecast DA, but that regional monitoring quality currently places the strongest upper bound on forecast skill.

The paper's main operational result is that magnitude forecasting and closure-warning are related but distinct tasks. Regression-only spike recall reaches 0.857 at the 12 ¯g/g threshold on the deterministic holdout, but thresholding the regression ensemble at 20 ¯g/g catches only a fraction of the below-to-above-threshold transitions that trigger harvest closures, because MSE-optimized regression conservatively under-predicts extremes (Section 4.5). A dedicated transition-focused classifier substantially improves transition recall, and the resulting hybrid alert (classifier $\cup$ regression trigger) achieves 87.6% recall on the full retrospective panel. DATect therefore contributes both a quantitative forecasting pipeline and a transition-focused early-warning layer that can complement the qualitative PNW HAB Bulletin, with the entire tuning workflow (per-site configurations, evaluation windows, classifier thresholds) reproducible end-to-end from a single source-of-truth JSON file.

Author Contributions: Conceptualization, A.C.; methodology, A.C.; software, A.C.; validation, A.C.; formal analysis, A.C.; investigation, A.C.; data curation, A.C.; writing—original draft preparation, A.C.; writing—review and editing, A.C. and J.M.; visualization, A.C.; supervision, J.M. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The DATect source code and web dashboard are available at https://github.com/ansoncchen/DATect-Forecasting-Domoic-Acid. Processed datasets and pre-computed cache files are available upon reasonable request.

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Conflicts of Interest: The authors declare no conflict of interest.

Appendix A Per-Site Model Configuration

Table A1 summarizes the site-wise model selection, clipping rules, and development-set performance tiers used during configuration selection.

Table A1. Per-site ensemble configuration showing model selection, prediction clipping, and development-set performance tier used during configuration selection (see Table 6 for held-out test-set R^2).

Site	w_{XGB}	w_{RF}	Clip q	Clip max	Tier
Copalis	0.00	1.00	0.97	—	High
Twin Harbors	0.00	1.00	0.98	—	High
Quinault	0.00	1.00	0.98	—	High
Long Beach	1.00	0.00	0.98	—	High
Kalaloch	0.00	1.00	0.95	80	High
Clatsop Bch	1.00	0.00	—	—	Mod.
Gold Beach	1.00	0.00	0.95	—	Mod.
Cannon Bch	0.00	1.00	0.95	80	Low
Coos Bay	0.00	1.00	0.97	—	Low
Newport	1.00	0.00	0.98	—	Low

Clip q: predictions clipped to this quantile of training targets. "—" = global default ($q = 0.99$) or no hard ceiling. Clip max in ¯g/g. Blended weights did not improve over single-model selection (Appendix J).

Appendix B Classification Details

Table A2. DA risk categories used in DATect classification.

Category	DA Range (¯g/g)	Interpretation
Low	0–5	Safe for harvest
Moderate	5–20	Monitor closely
High	20–40	Elevated toxicity; closures likely
Extreme	>40	Dangerous; immediate closure

Table A3. Per-category classification performance on the held-out test set ($n = 2{,}181$). The Low category achieves the highest reliability, while High and Extreme categories are limited by sample size. Overall accuracy 75.1% (macro-average precision 0.605, recall 0.581, F1 0.591).

Category	DA Range	N	Precision	Recall	F1
Low	0–5 ¯g/g	1,382	0.865	0.858	0.862
Moderate	5–20 ¯g/g	523	0.593	0.642	0.617
High	20–40 ¯g/g	168	0.456	0.435	0.445
Extreme	≥40 ¯g/g	108	0.506	0.389	0.440
Overall (macro)		**2,181**	**0.605**	**0.581**	**0.591**
Overall accuracy					75.1%

Table A4. Confusion matrix for four-category DA classification. Most errors occur between adjacent categories.

Actual \ Predicted	Low	Moderate	High	Extreme
Low	1183	142	27	30
Moderate	164	327	30	2
High	16	80	67	5
Extreme	15	22	32	39

Appendix C Component Ablation

Table A5. Component ablation study showing the contribution of each DATect design choice. Each row removes one component from the full system. Evaluated on the held-out paper sample (seed = 123, $n = 1{,}202$) under the leak-free grid-winner per-site configuration.

Configuration	R^2	ΔR^2	MAE	Notes
Full DATect (baseline)	0.211	—	6.60	Current grid-winner config
No per-site custom.	0.023	-0.187	6.71	Largest impact
No interp. training	0.186	-0.024	6.90	Data augmentation contribution
No obs-order lags	0.208	-0.003	6.70	Redundant with persistence at 1-wk horizon
No derived features	0.211	$+0.000$	6.60	Negligible standalone impact
No OAD features	0.208	-0.003	6.59	Sub-noise; offshore signal doesn't reach beach

Appendix D Per-Site Hyperparameter Configuration

Table A6 presents the full XGBoost hyperparameter configuration for each site.

Table A6. Per-site XGBoost hyperparameter overrides. Parameters not listed use global defaults ($n_{\text{est}} = 400$, depth $= 6$, lr $= 0.05$, subsample $= 0.85$, colsample_bytree $= 0.85$, colsample_bylevel $= 0.8$, $\alpha = 0.1$, $\lambda = 1.0$, $\gamma = 0.1$, min_child_wt $= 3$, tree_method $=$ hist).

Site	dep.	n_{est}	lr	mcw	α	λ	γ	ss	cs
Kalaloch	2	80	.02	12	2.0	10	2.0	.6	.6
Quinault	3	200	.03	7	0.3	2.0	0.3	.8	.8
Copalis	2	100	.03	10	1.0	5.0	1.0	.7	.7
Twin Harb.	3	150	.03	8	0.5	3.0	0.5	.8	.8
Long Beach	3	250	.03	7	0.3	2.0	0.3	.8	.8
Clatsop Bch					*(global defaults)*				
Cannon Bch	2	80	.02	12	1.5	7.0	1.5	.6	.6
Newport	2	100	.02	15	2.0	10	2.0	.6	.6
Coos Bay	2	80	.02	15	2.0	10	2.0	.6	.6
Gold Beach	2	100	.02	15	2.0	10	2.0	.6	.6

mcw = min_child_weight; ss = subsample; cs = colsample_bytree. Clatsop Beach uses all global defaults.

Sites with low historical skill (Cannon Beach, Coos Bay, Gold Beach, Newport) use the conservative Random Forest configuration: $n_{\text{est}} = 200$, max depth $= 6$, min_samples_split $= 10$, min_samples_leaf $= 5$, max_features $= 0.5$.

Appendix E Feature Subsets by Site

Table A7. Feature category inclusion by site. Sites with "All" use the full feature set without subsetting. "partial" indicates a subset of the group (e.g., only `modis-sst` and `pdo` from Env. Core). Newport, Coos Bay, and Gold Beach use a single lag feature (`da_raw_prev_obs_1`) rather than a complete lag group; Cannon Beach uses the first two obs lags only (`obs_1`, `obs_2`). These sites were trimmed to $\leq$7 features to reduce overfitting given small training N. Features explored but removed from the pipeline are listed in Appendix F.

Site	Persist.	Full Lag	Short Lag	Full Roll	Short Roll	Env. Core	Climate	Disch.	PN
Kalaloch	✓		✓				✓	✓	✓
Quinault	✓	✓			✓	✓	✓	✓	
Copalis	✓		✓		✓		✓	✓	
Twin Harb.	✓	✓			✓	partial	✓	✓	
Long Beach	✓	✓		✓		✓	✓	✓	
Clatsop Bch					*All (no subset)*				
Cannon Bch	✓		✓			partial			
Newport	✓					partial			
Coos Bay	✓					partial			
Gold Beach	✓					partial			

Appendix F Features Explored but Removed

During development, we explored a number of additional features motivated by domain knowledge of harmful algal bloom dynamics. Individual feature ablation (Table A8) confirmed that none contributed more than $|\Delta R^2| = 0.005$ to ensemble performance, and all have been removed from the final pipeline. We document them here for the benefit of future researchers.

Derived Environmental Features

- **Marine heatwave flag**: Binary indicator for SST anomaly $> 1.5\,^\circ$C, motivated by the 2015 "Blob" event [6,7]. $\Delta R^2 = -0.004$.
- **BEUTI nonlinear terms**: BEUTI^2 for the upwelling "Goldilocks zone" [8] and a relaxation indicator for onshore transport [12]. $\Delta R^2 = -0.001$ and -0.002.
- **PDO–ONI phase coherence**: Binary flag for co-occurrence of positive PDO and El Niño [14]. $\Delta R^2 = -0.002$.
- ***Pseudo-nitzschia* threshold**: Binary indicator for cell counts $>$ 50,000 cells/L. $\Delta R^2 = -0.005$.
- **Fluorescence efficiency**: FLH / Chl-*a* ratio as a physiological stress proxy. $\Delta R^2 = -0.001$.
- **K490 quadratic**: K490^2 for turbidity nonlinearity. $\Delta R^2 = +0.001$ (removing *improves* performance).

Satellite Observations

Several satellite products were downloaded and tested but contributed negligibly:

- **MODIS chlorophyll-*a*** and **chlorophyll-*a* anomaly**: $\Delta R^2 = -0.004$ and -0.001, respectively.
- **MODIS K490** (diffuse attenuation coefficient): $\Delta R^2 = +0.001$ (removing improves performance).
- **MODIS PAR** (photosynthetically active radiation): $<$ 1% feature importance.

Removing all six satellite-derived features simultaneously costs only $\Delta R^2 = -0.003$.

Temporal and Other Features

Additional temporal encodings (`quarter`, `is_bloom_season`, sine/cosine of week-of-year, `days_since_start`), a recency feature (`weeks_since_last_raw`), site coordinates (`lat`, `lon`), and rolling-mean features for 8- and 12-week windows all had $<$ 1% feature importance and were removed.

Appendix G Web Application Details

Architecture and Deployment

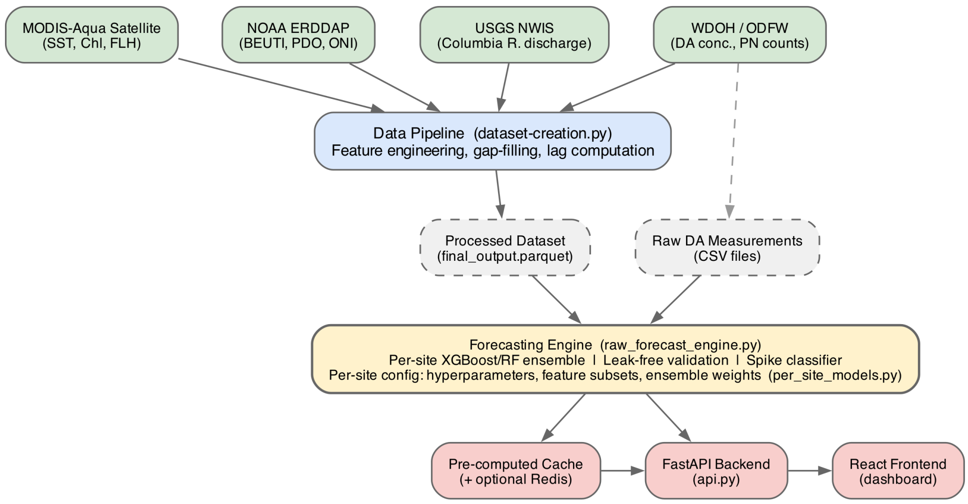

Figure A1. DATect system architecture. Data from satellite, climate, hydrological, and biological sources are processed into a unified dataset. Raw DA measurements are also read directly by the forecasting engine for raw-only persistence features. Results are served through a FastAPI backend to a React dashboard, with a pre-computed cache accelerating retrospective evaluations.

The DATect backend serves forecast requests, retrospective evaluation results, and visualization data (correlation heatmaps, sensitivity analyses, spectral analyses, feature importance rankings). A pre-computed cache system stores results from computationally intensive retrospective evaluations as JSON and Parquet files, with an optional Redis backend providing 100× faster cache reads for production deployments. The retrospective evaluation cache—encompassing all 2,181 validation points across 10 sites and 8 model configurations—was pre-computed on the University of Washington Hyak high-performance computing cluster (*klone* partition, STF allocation). Because each test point requires training a fresh ensemble on its own chronological training subset, the per-anchor computation is embarrassingly parallel and well-suited to HPC execution; running the full retrospective evaluation sequentially on a laptop would require several days, whereas the Hyak cluster completes the computation in approximately 2–3 hours across parallel jobs.

The system is containerized with Docker and deployed on Google Cloud Run, a serverless platform that scales to zero when idle and scales up to three concurrent instances under load. Cold start time is approximately 10–15 seconds. The deployment pipeline uses Google Cloud Build for continuous integration.

Operational Modes

The dashboard supports two modes. In **real-time forecasting mode**, users select a target date and monitoring site; DATect trains a fresh model on all data up to the anchor date, returning a point prediction, 90% confidence interval, risk category, and per-category probability distribution. In **retrospective analysis mode**, users select a model type and task to browse pre-computed validation results across all 2,181 test points, with interactive Plotly charts for scatter plots, confusion matrices, residual distributions, and time series overlays, filterable by site, date range, and DA risk category. Seven additional visualization types—including correlation heatmaps, sensitivity analyses, spectral density plots, SHAP-style feature waterfalls, and an interactive geographic map—are available for exploratory analysis. The live interface is available at the repository URL listed in the Data Availability Statement.

Appendix H Individual Feature Ablation

Table A8 presents a fine-grained ablation in which each row removes one feature or feature group from the full system. All changes fall within sampling variability ($|\Delta R^2| \leq 0.005$), confirming negligible marginal contributions. All features listed were removed from the final pipeline; see Appendix F for documentation.

Table A8. Feature group ablation. Each row removes one feature or group from the full system. All changes fall within sampling variability ($|\Delta R^2| \leq 0.005$), confirming negligible marginal contributions. Evaluated on the development set (seed = 42) under an earlier per-site configuration (baseline $R^2 = 0.457$); the component ablation table (Table A5) uses the final configuration ($R^2 = 0.414$). Since all $|\Delta R^2| \leq 0.005$, this conclusion is robust to the configuration difference.

Feature(s) removed	R^2	ΔR^2	MAE	ΔMAE
Full DATect (baseline)	0.457	—	6.15	—
All satellite (6)	0.454	-0.003	6.19	$+0.03$
All derived env. (4)‡	0.455	-0.002	6.18	$+0.03$
PN features (2)	0.452	-0.005	6.22	$+0.06$
Rolling std (3)	0.457	-0.001	6.16	$+0.00$
All low-imp. combined (8)†	0.453	-0.004	6.19	$+0.04$

†Includes: K490, K490^2, fluor. efficiency, Chl-*a* anomaly, PDO–ONI phase, BEUTI relaxation, PN threshold, Chl-*a*. ‡4 derived features active at this checkpoint; full set of 7 in Appendix F.

Appendix I Feature Importance Analysis

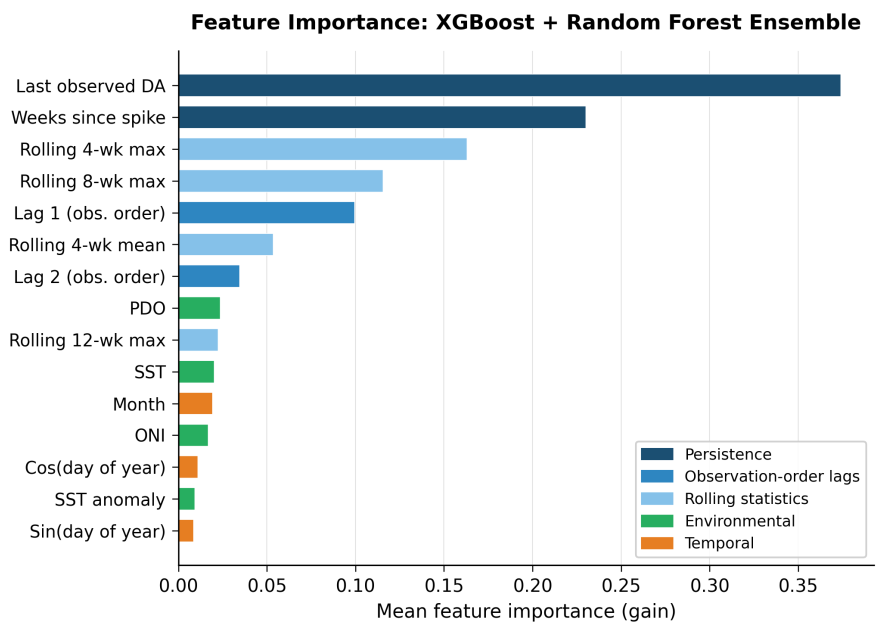

Figure A2. Mean feature importance (gain) across all development-set predictions (n = 1,177), averaged over per-site ensemble models. Features are color-coded by category. Persistence features dominate, consistent with strong DA autocorrelation at weekly timescales; environmental features contribute meaningful but secondary signal. See Section 4.6 for interpretation.

Appendix J Model Selection Robustness

Since DATect assigns each site a single model (XGBoost or Random Forest) based on development-set performance, weight robustness reduces to model-selection robustness: how sensitive are aggregate results to the per-site XGB vs. RF choice? Table A9 compares three strategies evaluated on the held-out test set ($n = 2{,}181$, seed = 123).

Table A9. Model selection robustness on the held-out test set ($n = 2{,}181$, seed = 123). The oracle selects the better single model per site using test-set R^2 directly. Bootstrap 95% CIs computed with $B = 10{,}000$ resamples.

Strategy	R^2	**MAE (¯g/g)**	**RMSE (¯g/g)**
XGBoost at all sites	0.276	6.58	17.08
RF at all sites	0.195	6.46	18.02
Per-site dev-set selection (this paper)	0.238	6.53	17.52
Linear/Ridge at all sites	0.203	6.61	17.93

The near-equivalence of these strategies ($R^2 = 0.195$–0.276, all within their pairwise bootstrap CIs) confirms that model-family selection is not a meaningful source of variance. Per-site dev-set selection (the ensemble formulation used throughout this paper) sits between XGBoost-everywhere (0.276) and RF-everywhere (0.195): on the held-out seed = 123 sample, XGB-everywhere narrowly wins, but this difference is well within the multi-seed 5-seed bootstrap dispersion ($\sigma \approx 0.08$ at this N; see Section 4.3). The ensemble formulation is retained for generality and because per-site selection enables site-specific feature subsets and clipping (Appendix D); choosing XGB-everywhere would slightly degrade per-site fit at WA sites where RF performs comparably with lower prediction variance.

Appendix K Perturbation Sensitivity Analysis

To validate that per-site design choices are not fragile artifacts of the seed-42 development set, we conduct a systematic perturbation study. Each experiment modifies a single design dimension while holding all others fixed, evaluated on the development set ($n = 1{,}177$, seed = 42, 20% sample). Table A10 reports the results.

Table A10. Perturbation sensitivity analysis. Each row modifies one design choice relative to the baseline configuration. ΔR^2 is relative to baseline ($R^2 = 0.409$). **Note on baseline discrepancy:** this differs from the component ablation baseline ($R^2 = 0.414$, Table A5) because the perturbation study was run on an intermediate configuration checkpoint that preceded the final monotonic-constraint and Oregon feature-reduction changes; both studies use the same development set (seed = 42, $n = 1{,}177$) and the ΔR^2 conclusions are unaffected.

Perturbation	R^2	ΔR^2	**MAE**
Model selection			
RF $\rightarrow$ XGB	0.381	−0.029	6.47
XGB $\rightarrow$ RF	0.405	−0.005	6.38
No per-site config	0.307	−0.103	6.66
RF hyperparameters			
Shallow (dep. 6, n_{est} 200)	0.410	+0.000	6.40
Deep (dep. 16, n_{est} 600)	0.409	−0.000	6.42
More trees (n_{est} 800)	0.410	+0.000	6.42
Less reg. (leaf = 1)	0.409	−0.001	6.43
Feature & clipping			
All features all sites	0.415	+0.006	6.45
Minimal features (4)	0.412	+0.002	6.66
Relax clip ($q = 0.99$)	0.380	−0.029	6.49
Disable clipping	0.384	−0.026	6.39
Disable monotonic	0.410	+0.000	6.43

Three findings emerge. First, Random Forest hyperparameters are genuinely insensitive: across four configurations spanning depth 6–16 and 200–800 estimators, no perturbation exceeds $|\Delta R^2| = 0.001$. This validates the claim in Section 4.6 that RF robustness to hyperparameters does not require per-site RF tuning. Second, per-site prediction clipping provides meaningful regularization: relaxing all per-site quantile thresholds from their current values (0.95–0.98) to a uniform 0.99 costs $\Delta R^2 = -0.029$; notably, disabling clipping entirely costs only -0.026, revealing that the harm is not from clipping itself but from replacing well-calibrated site-specific thresholds with an overly permissive uniform one. Third, feature subsets have negligible pooled impact (+0.006 with all features, +0.002 with only four persistence features), confirming that predictive skill is dominated by the DA persistence signal rather than feature engineering.

Multi-seed evaluation further characterizes the stability of these results. Table A11 reports per-site R^2 across five random seeds with per-site configuration enabled.

Table A11. Per-site R^2 across five random seeds (20% sample, per-site configuration enabled). Sites are grouped by region.

Site	s42	s123	s456	s789	s1337	Mean	Std
Washington							
Copalis	.752	.764	.769	.712	.733	.746	.021
Quinault	.491	.584	.677	.680	.519	.590	.078
Twin Harb.	.514	.566	.579	.709	.566	.587	.065
Long Beach	.643	.526	.605	.642	.516	.586	.055
Kalaloch	.618	.627	.454	.628	.537	.573	.069
Oregon							
Clatsop Bch	.374	.305	−.238	.666	.296	.280	.292
Coos Bay	.358	−.004	.095	−.001	.153	.120	.133
Cannon Bch	.014	−.071	.138	−.055	−.071	−.009	.080
Gold Beach	−.053	.035	−.251	.227	.006	−.007	.154
Newport	.119	−.163	−.950	−.113	−.226	−.267	.361
Pooled						.265	.102

Washington sites show stable performance across seeds (mean $R^2 = 0.57$–0.75, std ≤ 0.08), while Oregon sites exhibit high variance (std up to 0.36 for Newport). The regional skill divide persists across all five seeds, confirming that it reflects fundamental data quality differences rather than seed-specific tuning artifacts.

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